

Quiz 08 v1**Name:** _____*Essential Skills – Must show work for full credit.***1. Factor completely:**

$$6x^3 - 24x$$

2. Simplify:

$$\frac{x^2 - 36}{x - 6}$$

State any restrictions.

3. Simplify:

$$\frac{\frac{x^2-9}{x}}{\frac{x-3}{x}}$$

4. Divide:

$$\frac{x^3 - 4x^2 + x - 4}{x - 2}$$

5. Evaluate:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

Classify the discontinuity.

6. Evaluate the one-sided limits:

$$\lim_{x \rightarrow 0^-} \frac{1}{x} \quad \lim_{x \rightarrow 0^+} \frac{1}{x}$$

Does the two-sided limit exist? Explain.

Quiz 08 v2**Name:** _____*Essential Skills – Must show work for full credit.***1. Divide using long division:**

$$\frac{x^3 - 6x^2 + 5x - 30}{x - 6}$$

2. Factor completely:

$$4x^3 - 32x$$

3. Simplify:

$$\frac{\frac{x^2 - 25}{x}}{\frac{x - 5}{x}}$$

4. Reduce and state restrictions:

$$\frac{x^2 - 64}{x - 8}$$

5. Determine whether the following limit exists:

$$\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$$

If it exists, classify the type of discontinuity.

6. Evaluate both one-sided limits:

$$\lim_{x \rightarrow -2^-} \frac{1}{x + 2} \quad \lim_{x \rightarrow -2^+} \frac{1}{x + 2}$$

Explain whether a two-sided limit exists.

Quiz 08 v3**Name:** _____*Essential Skills – Must show work for full credit.***1. Simplify and state restrictions:**

$$\frac{x^2 - 100}{x - 10}$$

2. Divide:

$$\frac{x^3 - 2x^2 - 9x + 18}{x - 3}$$

3. Factor completely:

$$9x^3 - 81x$$

4. Simplify:

$$\frac{\frac{x^2 - 36}{x}}{\frac{x - 6}{x}}$$

5. Evaluate:

$$\lim_{x \rightarrow -4} \frac{x^2 - 16}{x + 4}$$

Classify the discontinuity.

6. Evaluate the one-sided limits:

$$\lim_{x \rightarrow 1^-} \frac{1}{x - 1} \quad \lim_{x \rightarrow 1^+} \frac{1}{x - 1}$$

Does the two-sided limit exist? Justify.

Quiz 08 v4**Name:** _____*Essential Skills – Must show work for full credit.***1. Use synthetic division:**

$$\frac{x^3 - 4x^2 - x + 4}{x - 2}$$

Write the remainder properly.

2. Factor completely:

$$12x^3 - 108x$$

3. Simplify and state restrictions:

$$\frac{x^2 - 144}{x - 12}$$

4. Simplify:

$$\frac{\frac{x^2 - 16}{x}}{\frac{x - 4}{x}}$$

5. Evaluate algebraically:

$$\lim_{x \rightarrow -3} \frac{x^2 - 9}{x + 3}$$

Describe what happens to the graph at that value.

6. A function is defined as:

$$f(x) = \begin{cases} x + 1 & x < 2 \\ 3 & x = 2 \\ x - 2 & x > 2 \end{cases}$$

Determine whether the limit exists as $x \rightarrow 2$ and justify your answer.

Quiz 08 v5**Name:** _____*Essential Skills – Must show work for full credit.***1. Simplify using the LCD method:**

$$\frac{\frac{1}{x} + \frac{2}{x-1}}{\frac{3}{x(x-1)}}$$

2. Divide using the box method:

$$\frac{3x^3 + 5x^2 - 6x - 10}{x + 2}$$

3. Add and simplify:

$$\frac{4}{x} + \frac{5}{x-3}$$

State all restrictions.

4. Factor completely using grouping:

$$6x^3 + 9x^2 - 4x - 6$$

5. Evaluate:

$$\lim_{x \rightarrow 1} \frac{1}{x - 1}$$

Describe the behavior of the function near $x = 1$.

6. Consider the rational function:

$$f(x) = \frac{x^2 - 4}{x - 2}$$

- a) Simplify the function.
- b) Identify the type of discontinuity.
- c) Explain what the graph looks like near that point.

Quiz 08 v6**Name:** _____*Essential Skills – Must show work for full credit.*

1. Use synthetic division and write the remainder properly:

$$\frac{x^3 - 2x^2 - 5x + 6}{x - 3}$$

2. Subtract and simplify:

$$\frac{6}{x} - \frac{2}{x - 4}$$

State all restrictions.

3. Simplify:

$$\frac{\frac{x^2 - 25}{x}}{\frac{x - 5}{x}}$$

4. Factor completely:

$$x^3 - 125$$

5. Evaluate:

$$\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$$

Classify the discontinuity and explain the graph behavior.

6. A function is defined by:

$$f(x) = \begin{cases} 2x - 1 & x < 3 \\ 5 & x = 3 \\ x + 4 & x > 3 \end{cases}$$

Determine whether $\lim_{x \rightarrow 3} f(x)$ exists and justify your answer.

Answer Key — Module 08 Quizzes

Quiz 08 v1

1. $6x^3 - 24x = 6x(x^2 - 4) = 6x(x - 2)(x + 2)$

2. $\frac{x^2-36}{x-6} = x + 6 \quad (x \neq 6)$

3. $x + 3 \quad (x \neq 0, 3)$

4. $x^2 - 2$ remainder 0

5. 4; removable discontinuity

6. $-\infty$ and $+\infty$; does not exist

Quiz 08 v2

1. $x^2 + 0x + 10$

2. $5x(x^2 - 4) = 5x(x - 2)(x + 2)$

3. $x + 5 \quad (x \neq 0, 5)$

4. $x + 7 \quad (x \neq 7)$

5. -6 ; removable discontinuity

6. $-\infty$ and $+\infty$; no two-sided limit

Quiz 08 v3

1. $x + 10 \quad (x \neq 10)$

2. $x^2 - x - 6$

3. $9x(x^2 - 9) = 9x(x - 3)(x + 3)$

4. $x + 6 \quad (x \neq 0, 6)$

5. -8 ; removable discontinuity

6. $-\infty$ and $+\infty$; does not exist

Quiz 08 v4

1. $x^2 - 2x - 1$ remainder 2

2. $12x(x^2 - 9) = 12x(x - 3)(x + 3)$

3. $x + 12 \quad (x \neq 12)$

4. $x + 4 \quad (x \neq 0, 4)$

5. -6 ; removable discontinuity

6. Jump discontinuity; limit does not exist

Quiz 08 v5

1. $\frac{x^2-x+1}{3}$

2. $3x^2 - x - 5$

3. $\frac{4(x-3)+5x}{x(x-3)} = \frac{9x-12}{x(x-3)}$

4. $(3x^2 - 2)(2x + 3)$

5. Infinite limit; vertical asymptote at $x = 1$

6. $\frac{x^2-4}{x-2} = x + 2$; removable discontinuity at $x = 2$

Quiz 08 v6

1. $x^2 + x - 2$ remainder 0

2. $\frac{6(x-4)-2x}{x(x-4)} = \frac{4x-24}{x(x-4)}$

3. $x + 5 \quad (x \neq 0, 5)$

4. $(x - 5)(x^2 + 5x + 25)$

5. 8; removable discontinuity

6. Jump discontinuity; limit does not exist

Module 08 Quiz Rubric

Overall Structure

- Questions 1–4: Essential Skills
 - Questions 5–6: Above & Beyond
 - Work must be shown for full credit.
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Essential Skills (Questions 1–4)

Grade	Description
A	Complete understanding demonstrated. All structural steps shown. Factoring is complete. Restrictions identified. Division process clear. Minor computational error does not reduce mastery level.
B	Strong understanding with one significant procedural error OR missing structural element. Work mostly clear and logically organized.
C	Partial understanding. Correct method identified but multiple execution errors OR incomplete reasoning.
D	Limited understanding. Attempt made but structural reasoning is weak or inconsistent.
Not Yet	Little to no evidence of understanding. Work missing or unrelated.

Above & Beyond (Questions 5–6)

Grade	Description
A	Correct computation AND precise mathematical explanation. Uses correct terminology (removable, infinite, jump, asymptote). Clear logical reasoning.
B	Correct conclusion with explanation that lacks precision or full justification.
C	Correct direction but incomplete reasoning or terminology errors.
D	Minimal explanation; reasoning unclear or inconsistent.
Not Yet	No meaningful explanation or incorrect conceptual reasoning.

Professional Expectations

- All work must be shown.
- Factoring must be complete.
- Restrictions must be stated clearly.
- Division remainders must be written correctly.
- Limit classifications must include reasoning.
- Mathematical notation must be precise.